

Lesson 9

Conditional Expectation

"Conditional Probability Persisted"

Remember that the conditional pmf of Y given X is

$$P_{Y|X}(y|x) = \frac{P(X=x, Y=y)}{P(X=x)} = P(Y=y|X=x) \\ = \frac{P_{X,Y}(x,y)}{P_X(x)}$$

More generally, the conditional pmf of the discrete r.v.

Y on $(\Omega, \mathcal{F}, P)$ given $A \in \mathcal{F}$

is defined as:

$$P_{Y|A}(y) = P(Y=y|A) = \frac{P(Y=y \cap A)}{P(A)}$$

provided that $P(A) > 0$

Def: Assume that Y is a discrete r.v. on $(\Omega, \mathcal{F}, P)$ and $A \in \mathcal{F}$

and $P(A) > 0$. The conditional expectation of Y given A is defined as:

$$E[Y|A] = \sum_y y P_{Y|A}(y)$$

In particular, if A is the event $\{X=x\} = \{\omega \mid X(\omega) = x\}$

then $P_{Y|A}(y) = P_{Y|X}(y|x) = \frac{P(Y=y, X=x)}{P(X=x)}$

$$E[Y|A] = E[Y|X=x]$$

$$= \sum_y y P_{Y|X}(y|x)$$

The Law of The Unconscious Statistician (LOTUS)

for Conditional Expectation

If X and Y are discrete

r.v.'s, and $g: \mathbb{R} \rightarrow \mathbb{R}$

is a function, then

$$E[g(Y) | X=x]$$

$$= \sum_y g(y) p_{Y|X}(y|x)$$

The proof is similar to that of the basic LOTUS.

Example (p. 58, Lesson 7)

The joint pmf of X, Y is

given:

$$p_{X,Y}(x,y) = \begin{cases} \frac{2}{n(n+1)} \left(\frac{x}{x+1} \right)^y & \begin{matrix} x=0, \dots, n-1 \\ y=0, 1, 2, \dots \end{matrix} \\ 0 & \text{otherwise} \end{cases}$$

Find $E[Y | X=x]$.

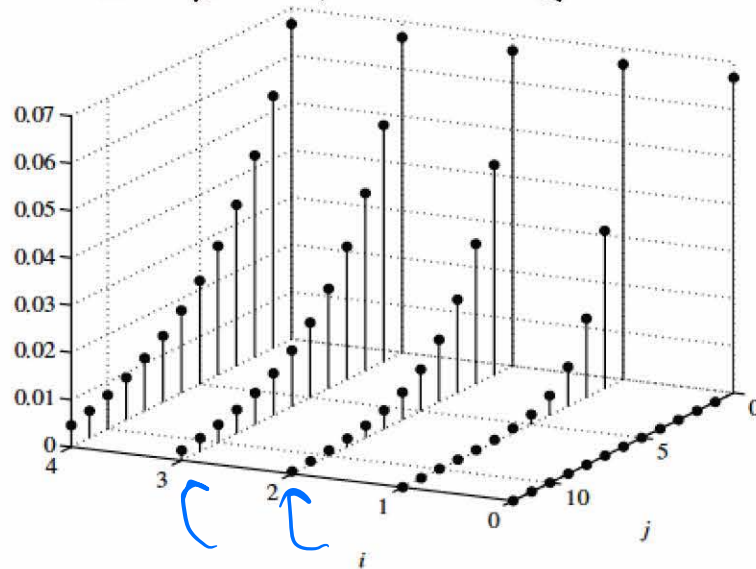


Figure 3.3. Sketch of bivariate probability mass function $p_{XY}(i, j)$ of Example 3.10 with $n = 5$. For fixed i , $p_{XY}(i, j)$ as a function of j is proportional to $p_{Y|X}(j|i)$, which is $\text{geometric}_0(i/(i+1))$. The special case $i = 0$ results in $p_{Y|X}(j|0) \sim \text{geometric}_0(0)$, which corresponds to a constant random variable that takes the value $j = 0$ with probability one.

We saw that $Y | \bar{X} = x \sim \text{Geo}_0\left(\frac{1}{x+1}\right)$

i.e.
$$P_{Y|\bar{X}}(y|x) = \left(\frac{1}{x+1}\right) \left(\frac{x}{x+1}\right)^y$$

$\underbrace{\qquad\qquad\qquad}_{p} \quad \underbrace{\qquad\qquad\qquad}_{(1-p)^y}$

Then $E[Y | X=x] =$

$$E[Y | X=x] = \frac{1 - \frac{1}{x+1}}{\frac{x+1-1}{x+1}} = \frac{1 - \frac{1}{x+1}}{\frac{x}{x+1}} = \frac{x+1-1}{x} = \frac{x}{x} = 1$$

A rectangular box with a decorative brown border and a notch at the top right corner. Inside, there are 12 horizontal lines for writing.

A rectangular box with a decorative brown border. Inside, there are 12 horizontal lines for writing.

Substitution Law for Conditional Expectation

Recall that

$$P(g(X, Y) = z \mid X = x)$$

$$= P(g(x, Y) = z \mid X = x)$$

The conditional law of the
unconscious statistician states
that

$$\mathbb{E}[g(X, Y) \mid X = x]$$

$$= \sum_y \left(\sum_{x'} g(x', y) P(X = x', Y = y \mid X = x) \right)$$

$$P(X=x', Y=y | X=x) = \frac{P(\overbrace{X=x', Y=y, X=x})}{P(X=x)}$$

$$A = \begin{cases} X=x, Y=y & x=x' \\ \emptyset & x \neq x' \end{cases}$$

$$P(X=x', Y=y | X=x) = \frac{P(X=x, Y=y)}{P(X=x)} \\ = P(Y=y | X=x) \leftarrow$$

$$E[g(X, Y) | X=x] = \sum_y g(x, y) P(Y=y | X=x) \\ = \sum_y g(x, y) P_{Y|X}(y|x) = E[g(x, Y) | X=x]$$

Therefore

$$E[g(X, Y) | X=x] = E[\underbrace{g(x, Y)}_{\downarrow} | \underbrace{X=x}_{\square}]$$

which is the substitution law for

Conditional expectation.

Exercise: If $g(x, y) = f(x)h(y)$,

what is $E[g(X, Y) | X = x]$?

$$\begin{aligned} E[g(X, Y) | X = x] &= E[f(X)h(Y) | X = x] \\ &= E[f(x)h(Y) | X = x] = f(x) E[h(Y) | X = x] \end{aligned}$$

The Law of Total Expectation

We saw that $E[g(X, Y) | X = x]$

is a function of x , $e(x)$.

We can use LOTUS to

calculate $E[e(X)]$:

$$\mathbb{E}[e(X)] = \sum_x e(x) P_X(x)$$

$$\text{but } e(x) = \mathbb{E}[g(X, Y) | X=x]$$

$$e(x) = \sum_y g(x, y) P_{Y|X}(y|x)$$

therefore,

$$\sum_x e(x) P_X(x) = \sum_x \left[\sum_y g(x, y) P_{Y|X}(y|x) \right] P_X(x)$$

$$= \sum_x \sum_y g(x, y) \underbrace{P_{Y|X}(y|x) P_X(x)}_{P_{X,Y}(x, y)}$$

$$= \sum_x \sum_y g(x, y) P_{X,Y}(x, y) = \mathbb{E}_{X,Y}[g(X, Y)]$$

$$\mathbb{E}_{X,Y}[g(X, Y)] = \sum_x \mathbb{E}_{Y|X}[g(Y, X) | X=x] P_X(x)$$

That means

$$E[g(X, Y)] = \sum_x E[g(X, Y) | X=x] P_X(x)$$

$\underbrace{\hspace{10em}}_{e(x)}$

=

$$E[e(X)]$$

where

$$e(x) = E[g(X, Y) | X=x]$$

If g is only a function of

Y :

$$E[g(Y)] = \sum_x E[g(Y) | X=x] P_X(x)$$

Try to show this

Corollary:

$$\sum_x \mathbb{E}[Y|X=x] P_X(x) = \mathbb{E}[Y]$$

Proof: $g(Y) = Y$ in the law of total expectation.

$$\text{If } g(Y) = \mathbb{E}[Y|X] \Rightarrow P(Y=y) = \sum_x P(Y=y|X=x) P_X(x)$$

Example: Light of intensity λ

is directed at a photo multiplier

that generates X primaries,

where $X \sim \text{Pois}(\lambda)$. The

photo multiplier generates Y

secondaries, where given $X=x$,

$Y|X=x$

Y is $\text{Bin}(x+1, 1/2)$. Find the

$E[Y|X=x] = \frac{x+1}{2}$ expected number of secondaries

and the correlation between X

and Y , defined as $E[XY]$.

$$E[Y] = \sum_x E[Y|X=x] P_X(x)$$

$$= \sum_{x=0}^{\infty} \left(\frac{x+1}{2} \right) P_X(x) = E\left[\frac{X+1}{2}\right]$$

$$= \frac{1}{2} E[X] + \frac{1}{2} = \frac{10}{2} + \frac{1}{2}$$

$$E[XY] = \sum_{x=0}^{\infty} E[XY|X=x] P_X(x)$$

$$= \sum_{x=0}^{\infty} E[xY|X=x] P_X(x)$$

$$\begin{aligned}
 &= \sum_{x=0}^{\infty} x \mathbb{E}[Y|X=x] P_X(x) \\
 &= \sum_{x=0}^{\infty} \frac{x(x+1)}{2} P_X(x) = \mathbb{E}\left[\frac{X(X+1)}{2}\right] \\
 &= \mathbb{E}\left[\frac{X^2 + X}{2}\right] \\
 &\mathbb{E}[X] = \lambda \quad \mathbb{E}[X^2] = \text{Var}(X) + (\mathbb{E}[X])^2 \\
 &= \frac{\lambda^2 + \lambda + \lambda}{2} = \frac{\lambda^2 + 2\lambda}{2} = \lambda + \lambda^2
 \end{aligned}$$

Total Expectation for
 Partitions of Sample Space
 Assume that $\{B_i \mid i \in I\}$
 is a partition of the sample
 space Ω and I is countable.

Also, assume that $\forall i, B_i \in \mathcal{F}$.

Define

$$\omega \in B_i \Leftrightarrow X(\omega) = i$$

Then X is a r.v. (why?)

$$\begin{aligned} P_X(i) &= P(X = i) = P(\{\omega \mid X(\omega) = i\}) \\ &= P(B_i) \end{aligned}$$

X is a discrete r.v. on $(\Omega, \mathcal{F}, P)$.

$$E[Y \mid X = i] = E[Y \mid B_i]$$

Applying the law of total expectation.

$$E[Y] = \sum_i E[Y \mid X = i] P_X(i)$$

$$= \sum_i E[Y \mid B_i] P(B_i)$$

Example: Two manufacturer of light bulbs compete in Los Angeles.

M_1 's light bulbs work for an average of 6,000 hours and

M_2 's light bulbs work for

an average of 3,000 hours.

M_1 supplies 30% of the light bulbs in L.A. and M_2

supplies 70% of the light bulbs in L.A. What is

the average life of a
light bulb in L.A.?

$L = \text{Life Time of A light Bulb}$

$$E[L] = \underbrace{\$ [L|M_1]}_{6000} \underbrace{P(M_1)}_{.3} \\ \rightarrow \underbrace{\$ [L|M_2]}_{3000} \underbrace{P(M_2)}_{.7} = 3900$$

The Conditional Expectation

as A Random Variable

Def:

Let X and Y be two discrete r.v.'s

on $(\Omega, \mathcal{F}, \mathbb{P})$ and define

$$e: \mathbb{R} \rightarrow \mathbb{R}$$
$$e(x) = \mathbb{E}[Y | X=x]$$

$e(X)$ is a r.v.; which is

called the Conditional

Expectation of Y given X .

$$\mathbb{E}[Y | X].$$

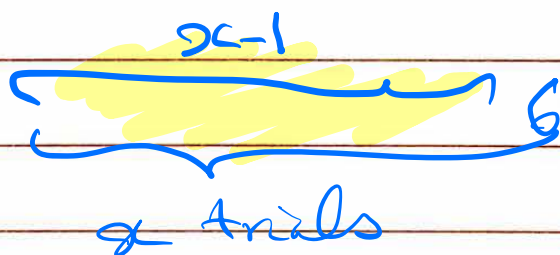
Example: Roll a die until
 a 6 is observed. Let X be
 the total number of the rolls,
 and let Y be the number
 of 1's that are observed.

Find $\mathbb{E}[Y|X=x]$ and

$\mathbb{E}[Y|X]$ and try to interpret

$\mathbb{E}[Y|X]$

$$X \sim \text{Geon}(\frac{1}{6})$$



$$Y|X=x \sim \text{Bin}(x-1, \frac{1}{6})$$

$$\mathbb{E}[Y|X=a] = \frac{a-1}{5} e(a)$$

$$\mathbb{E}[Y] = \mathbb{E}[Y|X] = e(X) = \frac{X-1}{5}$$

$$P_g(g) = P\left(\frac{X-1}{5} = g\right)$$

One interpretation:

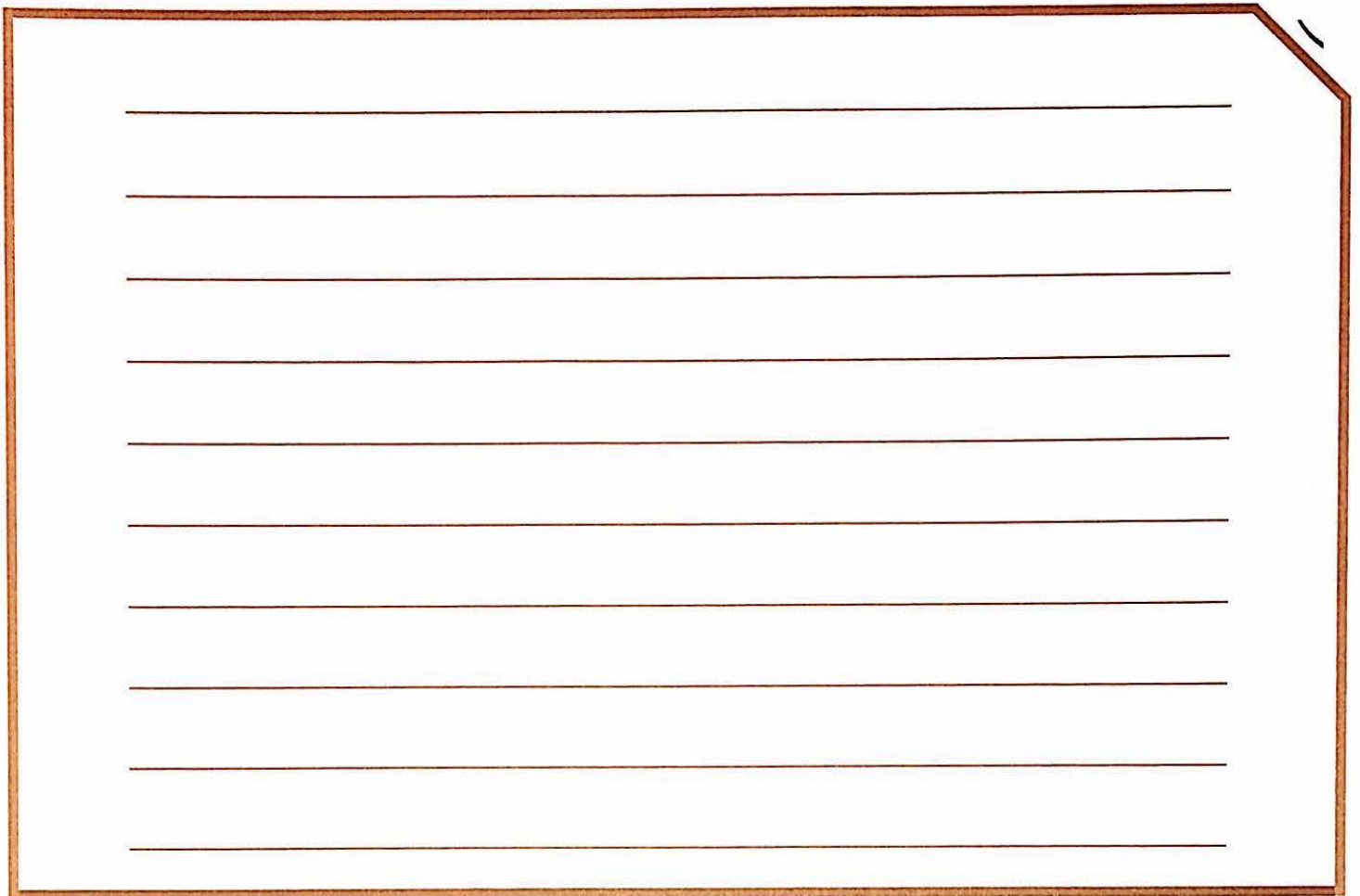
On average, the number of 1's

we see during this geometric trial is $\frac{\text{# of trials} - 1}{5}$

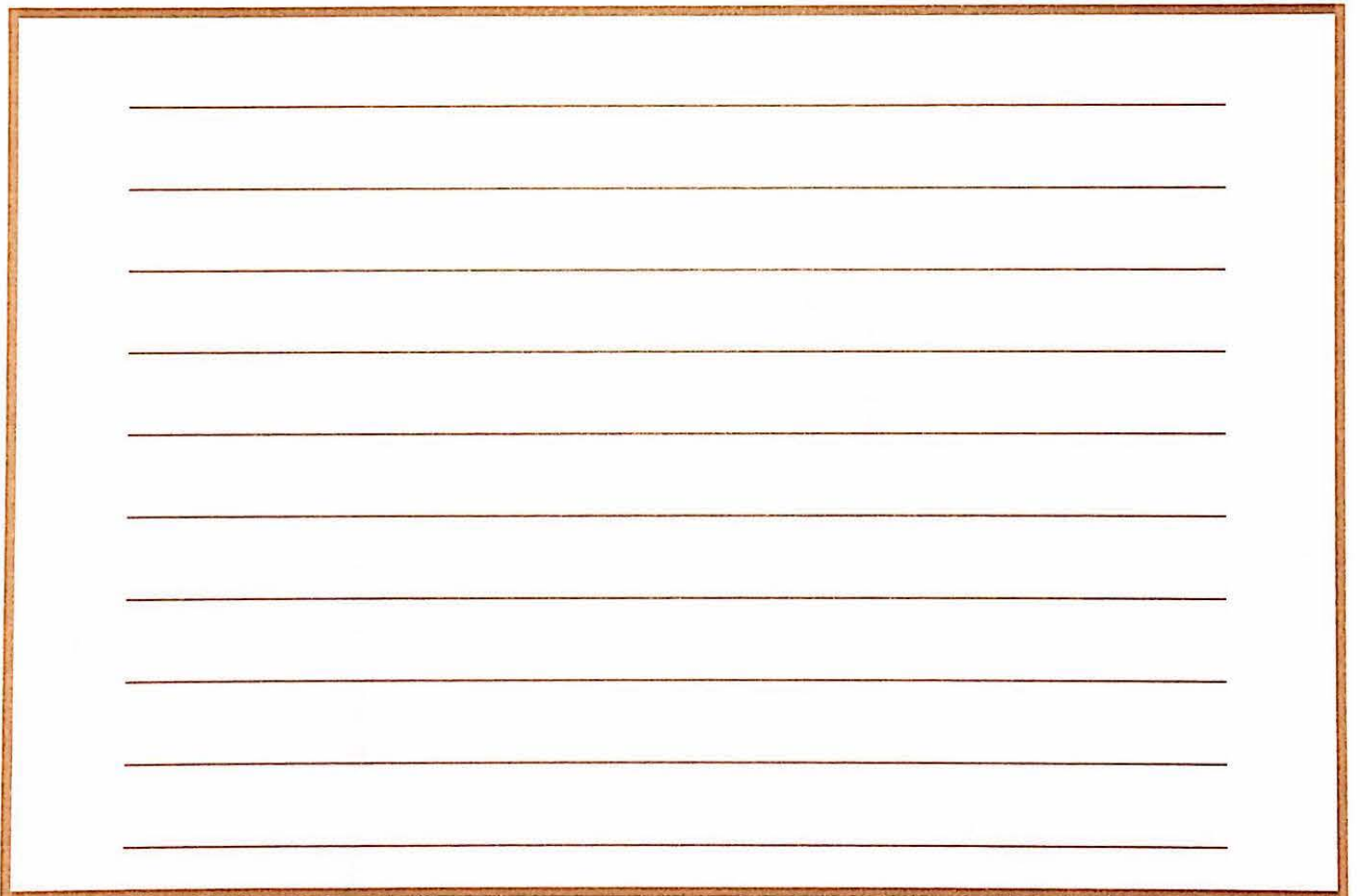
Further exercise:

Assume $X \sim \text{Uniform}(1, 6)$.

Calculate $\mathbb{E}[Y]$



A rectangular box with a brown border and a small notch in the top right corner. Inside, there are 12 horizontal brown lines for writing.



A rectangular box with a brown border. Inside, there are 12 horizontal brown lines for writing.

Conditioning on

Multiple Random Variables

Let X, Y, Z be discrete r.v.'s on $(\Omega, \mathcal{F}, \mathbb{P})$. Conditioning

on the event $A_z = \{\omega \mid \underline{X(\omega) = x} \} \cap \{\omega \mid \underline{Y(\omega) = y}\}$

results in

$$\mathbb{E}[Z \mid A] = \mathbb{E}[Z \mid \underline{Y=y}, \underline{X=x}]$$

$$\sum z \mathbb{P}_{Z \mid Y, X}(z \mid y, x) = \frac{\mathbb{P}(Z=z, Y=y, X=x)}{\mathbb{P}(X=x, Y=y)}$$

Denote $\mathbb{E}[Z \mid \underline{Y=y}, \underline{X=x}] = e(x, y)$.

Then

$e(X, Y)$ is a r.v. ~~that~~ that
is called the conditional
expectation of Z given
 X and Y and is denoted
as $E[Z | X, Y]$.

Remark: Because usually more than one r.v. is involved in calculating expectations and conditional expectations, sometimes a subscript is used to show the

distribution with respect to which the expectation is calculated:

$$\mathbb{E}_{Y|X}[g(X)h(Y)|X] = \sum_y g(x)h(y) P_{Y|X}(y|x)$$

$$\mathbb{E}_Y[g(X)h(Y)] = \sum_y g(x)h(y) P_Y(y)$$

$$\mathbb{E}_{X,Y}[g(X)h(Y)] = \sum_{x,y} g(x)h(y) P_{X,Y}(x,y)$$

Exercise: Calculate $E[Z | X=x, Y=y]$
using conditional pmfs.

$$E[Z | X=x, Y=y]$$

$$= \sum_{z \in \mathcal{Z}} z P_{Z|X,Y}(z|x,y)$$

Properties of Conditional

Expectation

Theorem:

Assume that X, Y , and Z

are ^{discrete} r.v.'s on $(\Omega, \mathcal{F}, P)$

and $a, b \in \mathbb{R}$ and $g: \mathbb{R} \rightarrow \mathbb{R}$.

$$a) \quad E[a|X] = a \quad \sum_y a P_{Y|X}(y|x) = a$$

$$b) \quad E[aY + bZ|X] = a E[Y|X] + b E[Z|X]$$

c) If $Y \geq 0$ a.s. then

$$E[Y|X] \geq 0 \quad \text{a.s.}$$

d) If X and Y are independent

$$E[Y|X] = E[Y]$$

(e) (The Tower Property) The Law of Iterated Expectations

$$E[E[Y|X]] = E[Y]$$

or in general

f) (Total Expectation)
(just a name change)

$$E_Y[g(Y)] = E_X[E_{Y|X}[g(Y)|X]]$$

Remark: Using the subscripts,

$$E_X[E_{Y|X}[Y|X]] = E_Y[Y]$$

and

$$E_X[E_{Y|X}[g(Y)|X]] = E_Y[g(Y)]$$

$$(g) E_{Y|X}[Y g(X) | X] = g(X) E_{Y|X}[Y | X]$$

Remark: (g) means that

the operator $E[\cdot | X]$ treats

functions of X as constants.

$$h) \mathbb{E}[g(X)|X] = g(X) \mathbb{E}[1|X]$$

$$(i) \mathbb{E}[Y|X, g(X)] = \mathbb{E}[Y|X]$$

Proof:

$$\begin{aligned} (f) \mathbb{E}[g(Y)] &= \sum_x \underbrace{\mathbb{E}[g(Y)|X=x]}_{\mathbb{E}[g(Y)|X]} P_X(x) \\ &= \mathbb{E}[\underbrace{\mathbb{E}[g(Y)|X]}_{\mathbb{E}[g(Y)|X]}] \end{aligned}$$

just by definition

(e) let $g(Y) = Y$ in (f)

$$\mathbb{E}[Y] = \mathbb{E}[\mathbb{E}[Y|X]]$$

$$(g) \mathbb{E}[Yg(X)|X] = g(X) \mathbb{E}[Y|X]$$

Let's focus on

$$\mathbb{E}[Yg(X)|X=x]$$

$$= \mathbb{E}[Yg(x)|X=x]$$

Substitution
Law

$$= \sum_y yg(x) P_{Y|X}(y|x) = g(x) \sum_y y P_{Y|X}(y|x)$$

$$= g(x) \mathbb{E}[Y|X=x]$$

$$\Rightarrow \mathbb{E}[g(X)Y|X] = g(X) \mathbb{E}[Y|X]$$

$$(h) \mathbb{E}[g(X)|X] = X$$

Use (g) by assuming $Y=1$

$$(i) \quad E[Y|X, g(X)] = E[Y|X]$$

$$E[Y | \underbrace{X=x, g(X)=z}_A]$$

$$A = \begin{cases} \emptyset & z \neq g(x) \\ X=x & z = g(x) \end{cases}$$

$$\rightarrow E[Y|X=x]$$

$$\Rightarrow E[Y|X, g(X)] = E[Y|X]$$

Exercise: Show the following property of conditional expectation

$$\mathbb{E} \left[\mathbb{E} [X | Y, Z] \mid Y \right]$$

$h(Y)$ $g(Y, Z)$

$$= \mathbb{E} [X | Y]$$

Note: It may be useful

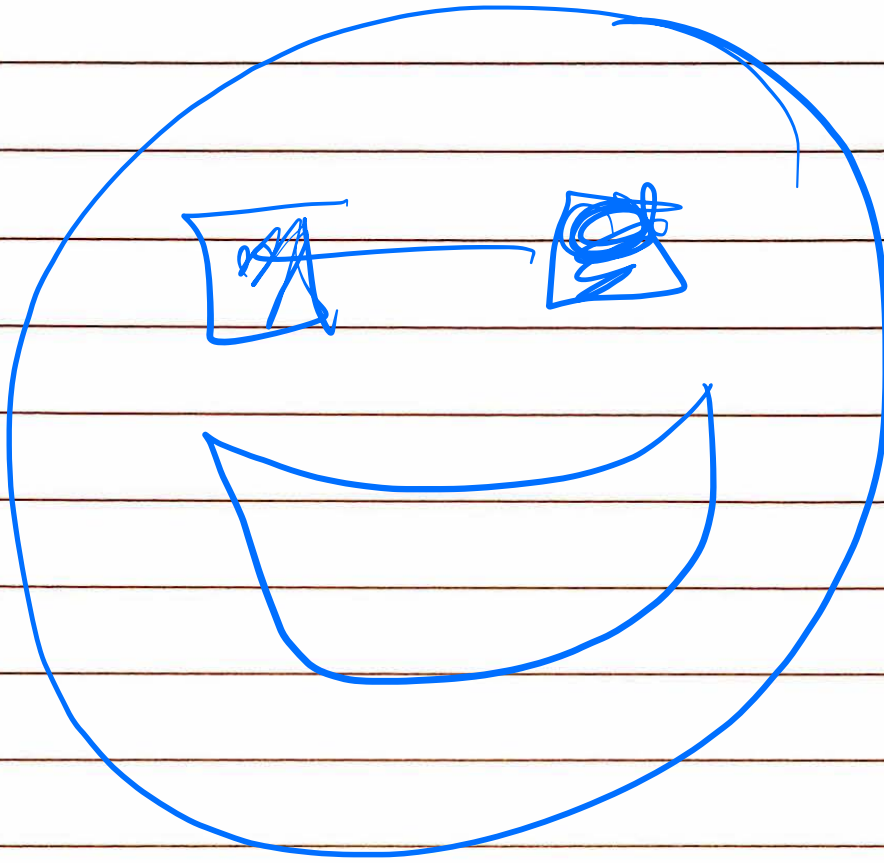
to calculate the RHS for

$$Y = y.$$

Note: You may want to check

that, intuitively, both RHS and

LHS are functions of Y .



Left to the Students

Existence of Conditional Expectation

Lemma: If X is a r.v. on $(\Omega, \mathcal{F}, \mathbb{P})$ and X is integrable, or non-negative,

$\mathbb{E}[|Y| | X=x]$ is finite,

for all x for which $P_X(x) \neq 0$

Proof:

and well-defined if Y is non-negative
 $\mathbb{E}[|Y| | X=x]$

$$\mathbb{E}[|Y| | X=x] = \sum_y |y| P_{Y|X}(y|x)$$

$$= \sum_y |y| \frac{P(x,y)}{P_X(x)}$$

$$\frac{1}{P_X(n)} \sum_y |y| P_{X,Y}(n, y) \leq \frac{1}{P_X(n)} \sum_y |y| P_Y(y) = \frac{1}{P_X(n)} E[|Y|] < \infty$$

Because $\{X=n \cap Y=y\} \subseteq \{Y=y\}$

$$P(\{X=n, Y=y\}) \leq P(Y=y)$$

Y is non-negative:

$$E[|Y| | X=a] \leq \frac{\sum_y |y| P_Y(y)}{P_X(a)}$$

$$= \frac{\sum_{y=0}^{\infty} y P_Y(y)}{P_X(a)}$$

Always well-defined

Y is non-negative $E[|Y| | X=a]$ well defined.

Remark: The converse is not
necessarily true, i.e. there
are cases for which
 $E[|Y| | X=x]$ is finite for
all x , but $E[|Y|] = \infty$.

Exercise: Find an example for the
 above Remark

Try:

$$E[|Y|] = \sum_x E[|Y| | X=x] P(x)$$

$\nwarrow$
infinite
 $\nwarrow$
finite

Integrable $\longrightarrow$ Conditionally integrable

The Reverse is not necessarily true

y is non-negative $\longrightarrow \mathbb{E}[|y| | X=x]$
well defined

Conditional Probability as Conditional Expectation

Recall that the probability

$$\mathbb{P}(X \in B) \text{ can be written as } \mathbb{E}[I_B(X)].$$

Similarly:

$$\mathbb{P}(Y \in B | X=x) = \mathbb{E}_{Y|X} [I_B(Y) | X=x]$$

$$\mathbb{P}(Y \in B | X) = \mathbb{E}_{Y|X} [I_B(Y) | X]$$

Proof: Left as an exercise for

X, Y discrete.

Remark: In fact, the concept of conditional expectation provides a rigorous foundation for defining probability. $E[\cdot | X]$ is more fundamental than $P(\cdot | X=x)$.
(c.f. Probability with Martingales by Williams)

Exercise (The Law of Iterated Expectation/ Total Expectation)

Assume that $X \sim \text{dU}(0, 5)$

and $Y|X=x \sim \text{Pois}(x+1)$.

Calculate $E[Y^2 X]$.

Naïve Solution = calculate

$P_{X,Y}$ and use LOTUS $\sum_{x,y} y^2 x P_{X,Y}(x,y)$

$$E[Y^2 X] = \sum_x E[Y^2 X | X=x] P_X(x)$$

$$= \sum_x E[Y^2 x | X=x] P_X(x)$$

$$= \sum_x x E[Y^2 | X=x] P_X(x) \quad \text{✗}$$

$$Y|X=x \sim \text{Pois}(x+1)$$

$$E[Y|X=x] = x+1$$

$$E[Y^2|X=x] = \text{Var}(Y|X=x) + (E(Y|X=x))^2 \\ = (x+1) + (x+1)^2$$

$$V = \sum_{x=0}^5 x [(x+1) + (x+1)^2] \underbrace{P_X(x)}_{1/6} \\ = \sum_{x=0}^5 x ((x+1) + (x+1)^2) / 6$$

Tower Property Solution ~

$$E_X [E[Y^2|X]] = E_{X,Y} [Y^2]$$

$$\text{LHS} = E_X [X E[Y^2|X]] = E_{X,Y} [Y^2]$$

$$E_X [X (X+1 + (X+1)^2)] \quad X \sim \text{Unif}(0,5)$$

Wald's Equality:

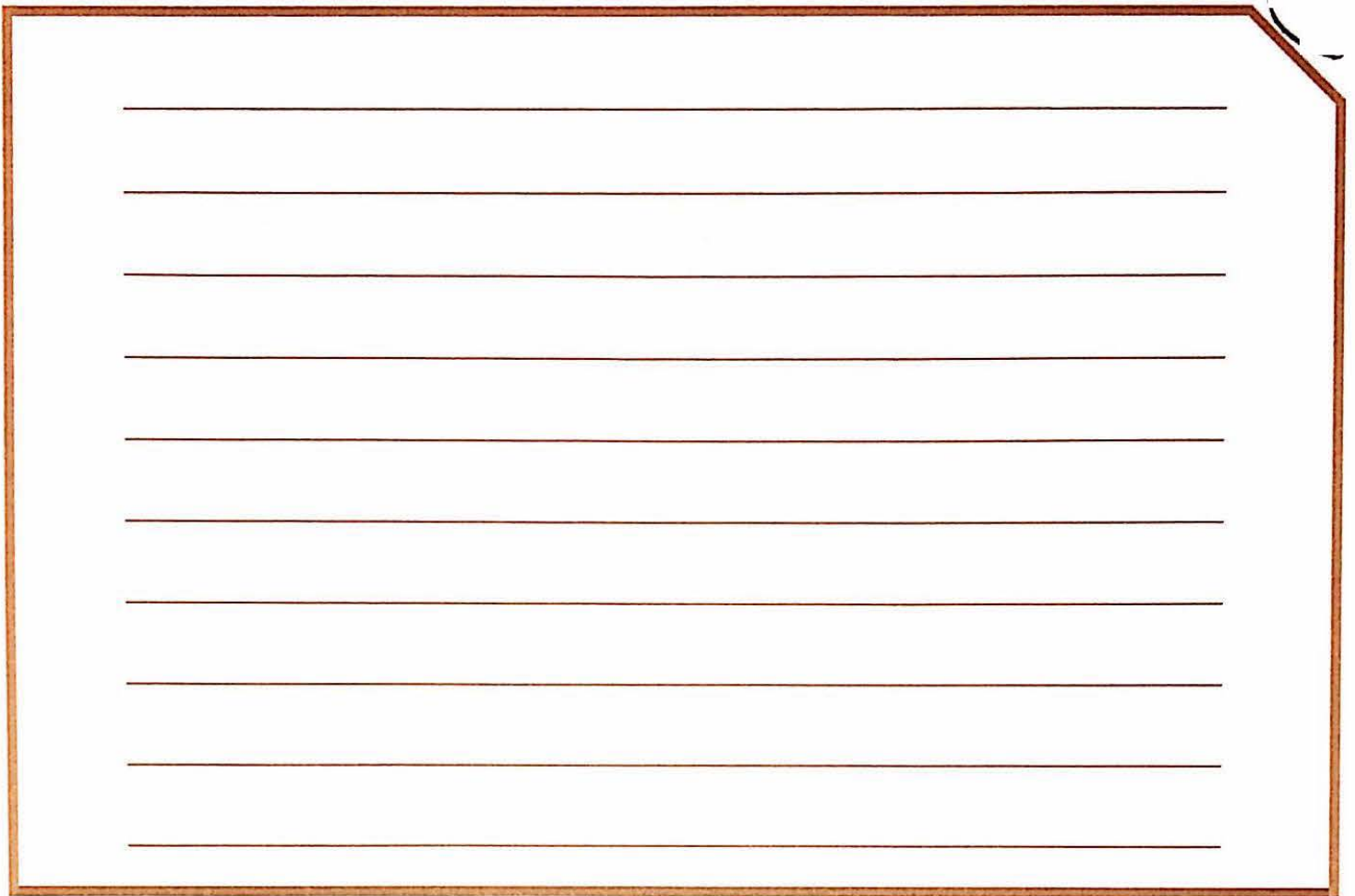
Let N be a random variable that takes non-negative integer values.

Let $X_1, X_2, \dots$ be a sequence of i.i.d discrete random variables

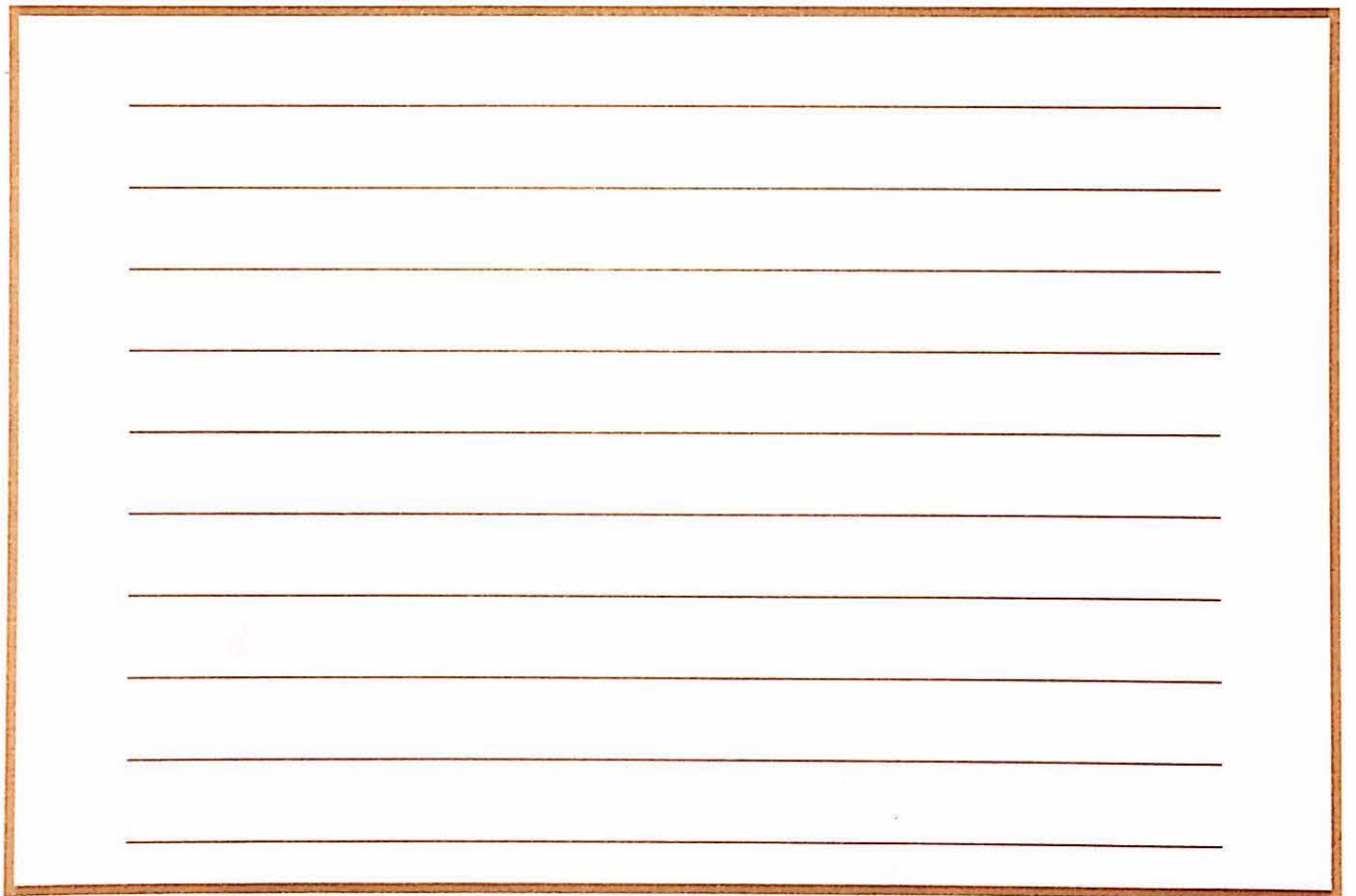
that have finite expectation and are independent from N .

Wald Equality states that

$$E \left[\sum_{i=1}^N X_i \right] =$$



A rectangular box with a brown border, containing ten horizontal lines for writing. The lines are evenly spaced and extend across the width of the box. The top right corner of the box is slightly rounded.



A rectangular box with a brown border, containing ten horizontal lines for writing. The lines are evenly spaced and extend across the width of the box. The bottom right corner of the box is slightly rounded.

Higher Order Conditional Moments

Def: For all $r \in \mathbb{N}$,

$E[Y^r | X]$ is called the

r^{th} moment of Y given X .
(Conditional)

Def: For all $r \in \mathbb{N}$,

$$E[(Y - E[Y | X])^r | X]$$

is called the r^{th} conditional
central moment of Y given X .

Remark: In particular

$$\text{Var}(Y|X) = E[(Y - E[Y|X])^2 | X]$$

is called the conditional variance
of Y given X

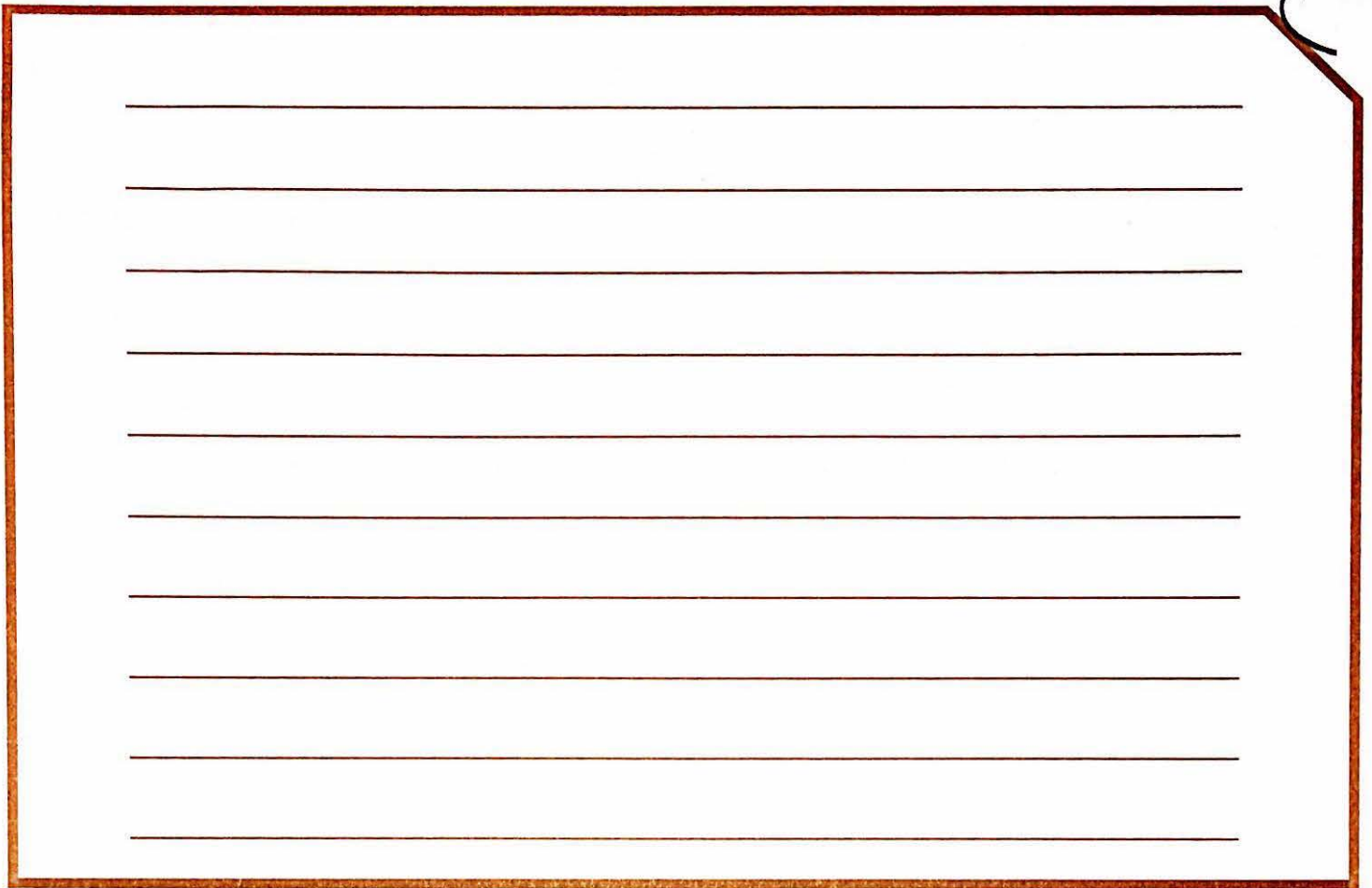
and $\sigma_{Y|X} = \sqrt{\text{Var}(Y|X)}$ is called

the conditional standard
deviation of Y given X

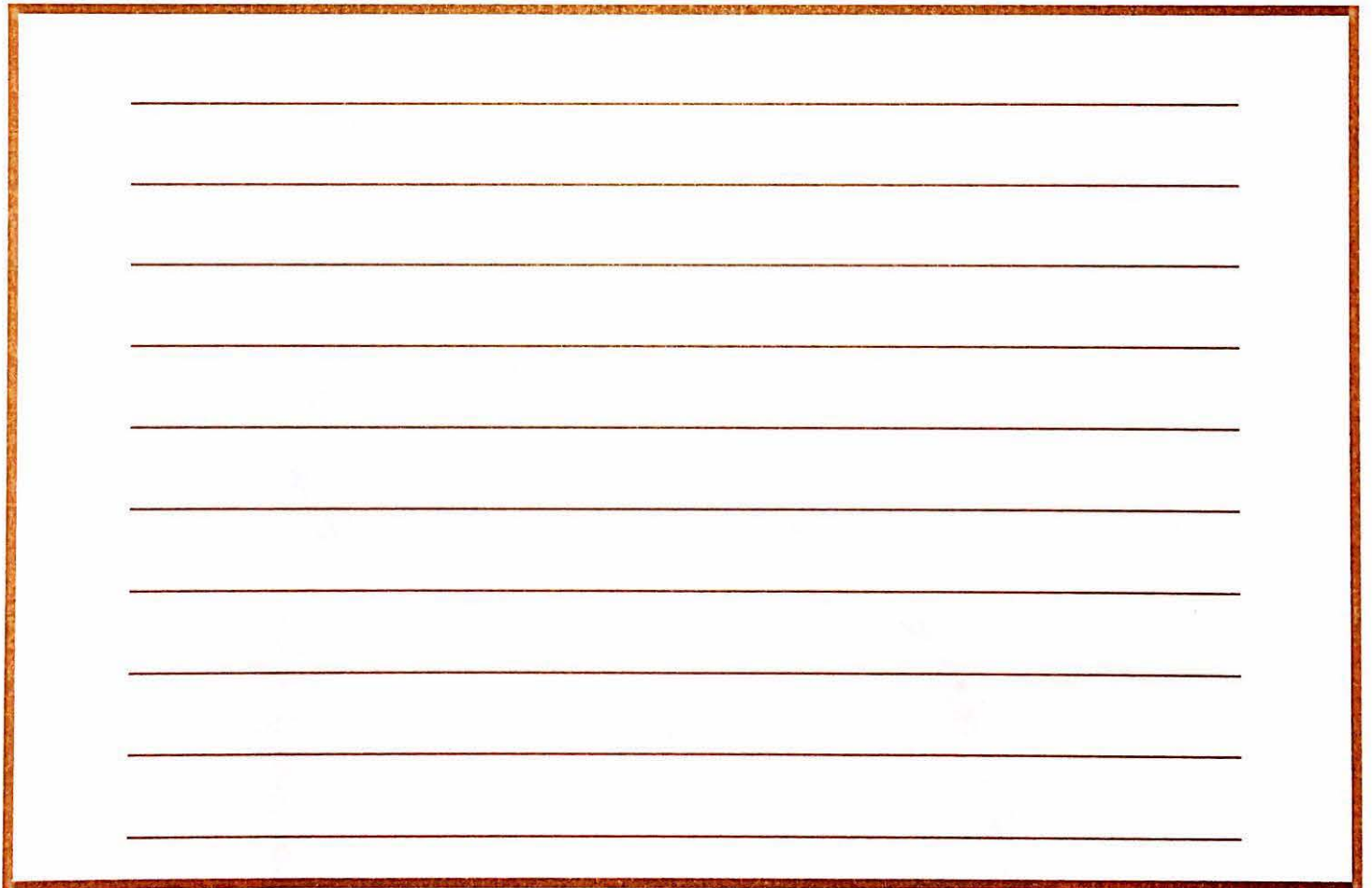
Remark: Conditional Variance inherits the properties of variance. For example:

Lemma) $\text{Var}(Y|X) = E[\text{Var}(Y|X)]$

Proof:



A rectangular box with a dark red border and a small notch in the top right corner. Inside the box are 12 horizontal lines for writing.



A rectangular box with a dark red border. Inside the box are 12 horizontal lines for writing.

Example: Roll a die until a 6 is observed. Let X be the total number of the rolls, and let Y be the number of 1's that are

observed. Find $\text{Var}(Y|X)$.

The image shows a blank page from a notebook or ledger. It features two distinct sections, each containing ten horizontal lines for writing. The lines are evenly spaced and extend across most of the width of each section. The entire page is enclosed within a decorative border that has a dark outer edge and a lighter, textured inner edge. The top right corner of the page is slightly folded over, and there is a small, dark mark near the top right corner of the first section.

Exercise : Let X and Y be independent random variables and $X \sim dU(0, 3)$ and $Y \sim dU(0, 3)$.

Let $Z = X + Y$. Find $E[Z|X]$, $E[X|Z]$, $E[XZ|X]$, $E[XZ|Z]$.

Projections, The Principle of Orthogonality, and The Projection Theorem

Def: Let C be a subset of the inner product vector space V .

For some $X \in V$, we call $\hat{X} \in C$ a projection of X onto C if

$$\|X - \hat{X}\| \leq \|X - Y\| \quad \forall Y \in C$$

Note: $\hat{X}$ can be viewed as an estimate of X , and $\|X - \hat{X}\|$

can be seen as the
estimation error,

The Principle of Orthogonality

Let S be a subspace of V .

(i.e. S is itself a vector space

and is a subset of V).

The "estimate" $\hat{x} \in S$ of $x \in V$

is ~~the~~^a minimum error estimate

of x , i.e.

$$\|x - \hat{x}\| \leq \|x - y\| \quad \forall y \in S$$

if and only if

$$\langle X - \hat{X}, Y \rangle = 0$$

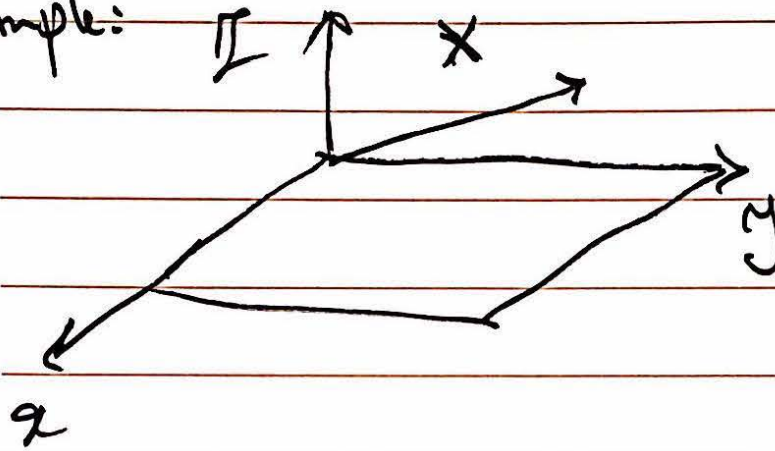
✓ $\forall Y \in S$

Remark: The principle of Orthogonality means for $\hat{X}$ an

estimate of X to be the "minimum error" estimate, the estimation error has to be orthogonal to the subspace in which the

estimate residues.

Example:



$$S = \left\{ \begin{bmatrix} x \\ y \\ z \end{bmatrix} \mid z=0 \right\}$$

S is a subspace

What is the "minimum error"

estimate of x , $\hat{x} \in S$?

Remark: The Principle of

Orthogonality does NOT

guarantee the existence of $\hat{x}$.

The Projection Theorem guarantees

the existence of $\hat{x}$.

Projection Theorem: If S is a "closed" subspace of V , and $X \in V$ then there exists a unique $\hat{X}$ such that

$$\|X - \hat{X}\| \leq \|X - Y\| \quad \forall Y \in S$$

Remark:

Def: S is called a closed set if any sequence X_n in S converges, it converges to a member of S , i.e. if

$\lim_{n \rightarrow \infty} \|X_n - X\| = 0$ then

$$X \in S.$$

Remark: One can see that we used the notion of "norm" to define convergence of sequences

more generally.

Conditional Expectation as An Estimator

Theorem: Let $g: \mathbb{R} \rightarrow \mathbb{R}$ be
a function, and $X, Y, g(X)$
random variables

$$\begin{aligned}\text{Then: } E[E[Yg(X)|X]] &= \\ E[E[X|X]g(X)] &= \\ E[Yg(X)]\end{aligned}$$

Remark: When $g(X) = 1$, we have
the tower property.

Proof

Remark: So far, we have
proved the results involving conditional
expectation and expectation
for discrete r.v.'s. Later on,
we see that most of them

are valid for other types of
random variables.

Interpretation :

$$E[E[Y|X]g(X)] = E[Yg(X)]$$

$$\Rightarrow E[(E[Y|X] - Y)g(X)] = 0$$

One can view the set of

all $g(X)$'s as a subspace

of the space of random

variables. Because $E[Y|X]$ is

a function of X , it is a

member of this subspace.

Let us think of $E[Y|X]$

as an estimate of Y , so

$\hat{Y} = E[Y|X]$, then:

$$E[(\hat{Y} - Y)g(X)] = 0$$

We learned that $E[Y_1 Y_2]$ is

the inner product between Y_1 and

Y_2 , therefore:

$$E[(\hat{Y} - Y)g(X)] = 0$$

$\Rightarrow$

$$\langle \hat{Y} - Y, g(X) \rangle = 0$$

In other words, the estimation error $\hat{Y} - Y$ is orthogonal to all possible $g(X)$'s, which consist a subspace.

By the projection theorem

$E[Y|X]$ is the minimum error estimate of Y in the subspace of $g(X)$'s, i.e.

$$E[(E[Y|X] - Y)^2] \leq E[(g(X) - Y)^2]$$

We can prove ~~this~~ for this particular case $\hat{Y} = E[Y|X]$. ^{it}

Theorem: Suppose that $E[X^2] < \infty$,
where X is a r.v. on $(\Omega, \mathcal{F}, \mathbb{P})$.
~~and~~ For any $g: \mathbb{R} \rightarrow \mathbb{R}$ for
which $g(X)$ is a random
variable

$$E[(Y - E[Y|X])^2] \leq E[(Y - g(X))^2]$$

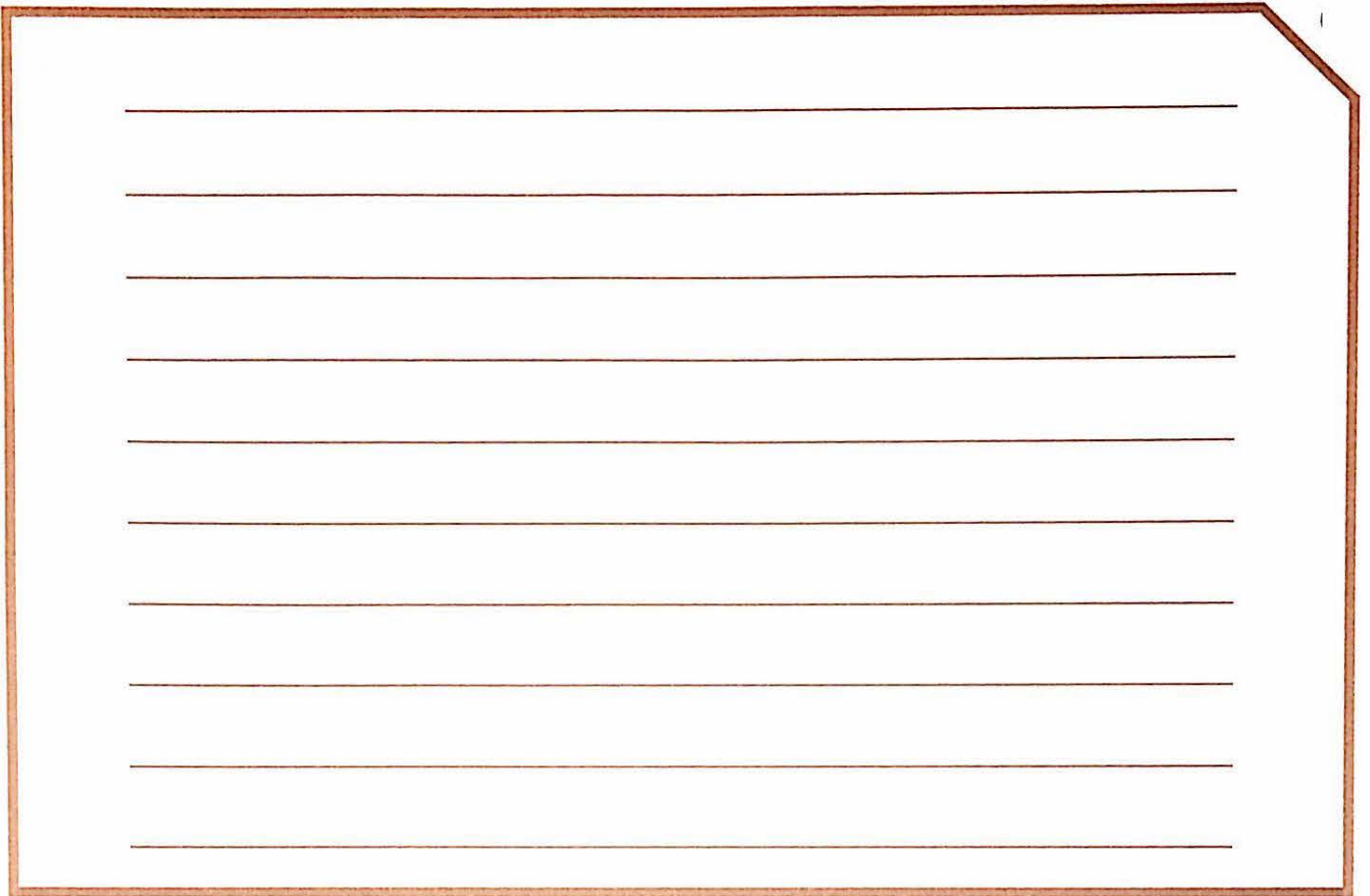
Remark: The right hand side of
the above equation is called
the Mean-Squared Error (MSE)
of the estimator $E[Y|X]$.

Consequently, $E[Y|X]$ is

called the Minimum Mean-Squared

^{Estimate}
Error (MSE) of Y .

Proof:



A rectangular box with a brown border and a notch in the top right corner. Inside the box, there are ten horizontal lines for writing, evenly spaced.



A rectangular box with a brown border. Inside the box, there are ten horizontal lines for writing, evenly spaced.

A More Rigorous Definition of Conditional Expectation

Definition: Assume that

$f: \Omega_1 \rightarrow \Omega_2$ is a function and

~~let~~ $(\Omega_1, \mathcal{F}_1)$ and $(\Omega_2, \mathcal{F}_2)$

are measurable spaces (i.e. $\mathcal{F}_1$

and $\mathcal{F}_2$ are σ -fields). We

say f is $\mathcal{F}_1, \mathcal{F}_2$ measurable

if the pre-image of any $B \in \mathcal{F}_2$

under $\mathcal{F}$ is a measurable set
in $\mathcal{F}_1$, i.e.

$$\forall B \in \mathcal{F}_2, f^{\leftarrow}(B) \in \mathcal{F}_1$$

Remark: A random variable

$$X: \Omega \rightarrow \mathbb{R}$$

is a $\mathcal{B}, \mathcal{F}$ measurable function,
because $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$ and $(\Omega, \mathcal{F})$
are measurable [^]
spaces

and

$$\forall B \in \mathcal{B}(\mathbb{R}), f^{\leftarrow}(B) \in \mathcal{F}.$$

Def. (Conditional Expectation of Y given X):

Assume that X, Y are r.v.'s on $(\Omega, \mathcal{F}, \mathbb{P})$. A random variable of the form $e(X)$,

where $e: \mathbb{R} \rightarrow \mathbb{R}$ is a measurable function is called a conditional expectation of Y given X iff for all measurable functions $g: \mathbb{R} \rightarrow \mathbb{R}$,

Note: #

(a) The existence of $e(X)$ is guaranteed if Y is integrable.
(We proved it for discrete r.v.'s)

(b) Any two functions e_1 and e_2 that satisfy the above definition are equal with probability one:

$$P(\{\omega \mid e_1(X(\omega)) = e_2(X(\omega))\}) = 1$$

Therefore, when using " $=$ " sign for conditional expectation, we must remember that we imply equality with probability one. In other words, Conditional

expectation of Y given X is actually a class of r.v.'s that are equal with probability 1.

For Further Discussion, see:

Probability with Martingales, by Williams.